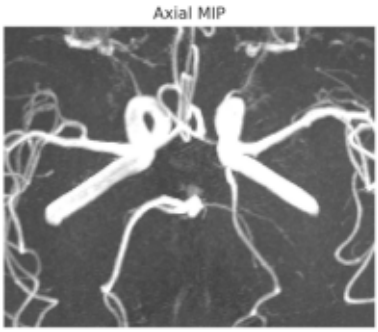
**a** Source MRA




**b** Given vessel foreground

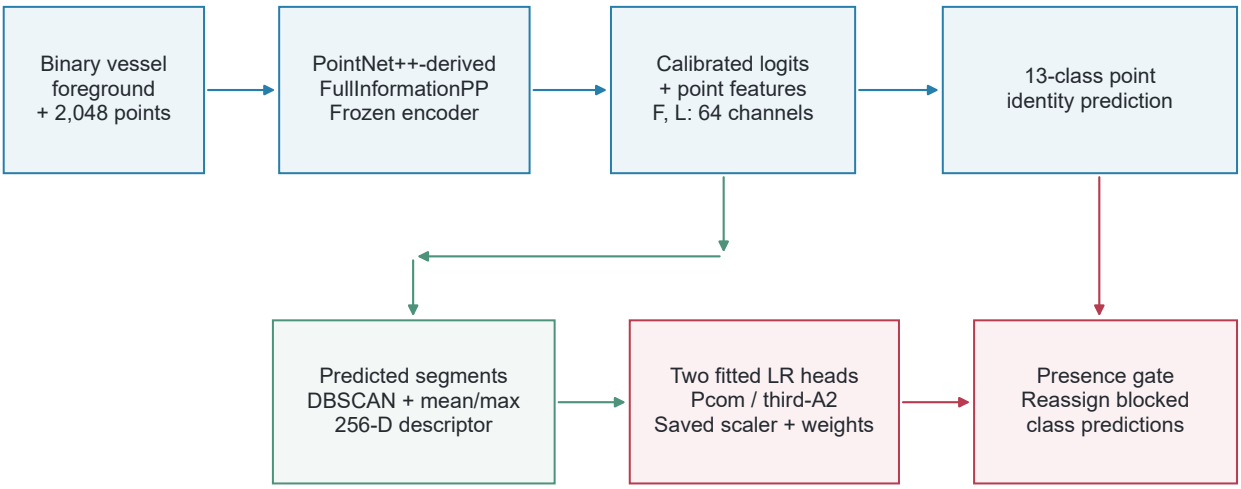


**c** Reference identity colors



BA L-PCA R-MCA L-MCA L-Pcom R-ACA 3rd-A2
R-PCA R-ICA L-ICA R-Pcom Acom L-ACA

**d** StrokePoint++-Gate: implemented workflow




Given-foreground anatomical labeling. No image-to-vessel extraction or stroke-risk prediction.

**e** Separate evidence roles and versions

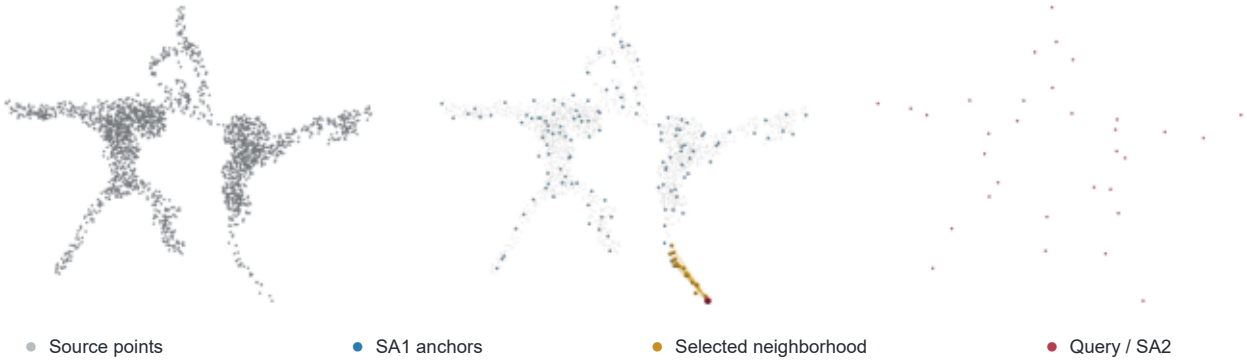
| Original application | Single-MRA reconstruction | External fixed evaluation |
|---|---|---|
| 48 fit / 65 internal patients<br>Paired CT/MR aggregation | Fixed R0v2 recipe<br>No recovery of original LR states | Lausanne DEV: 20<br>IXI-HH FINAL: 20 |

Deployment geometry is label-blind; reference identity colors and source MRA are shown for display, not as Gate inputs.

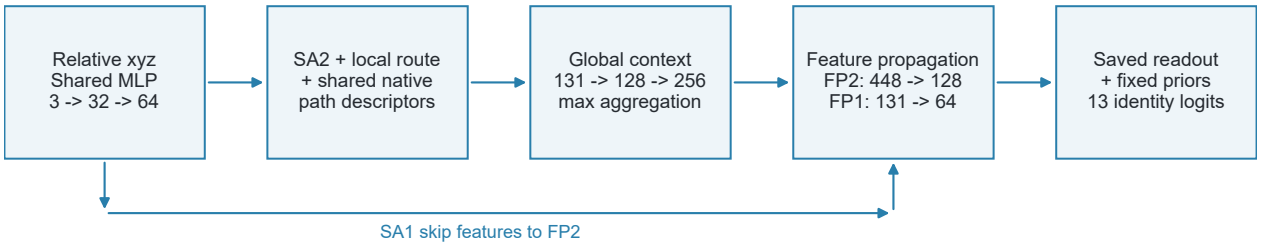
**a** Input: 2,048 points

**b** SA1: 128 anchors + grouping

**c** SA2: 32 anchors



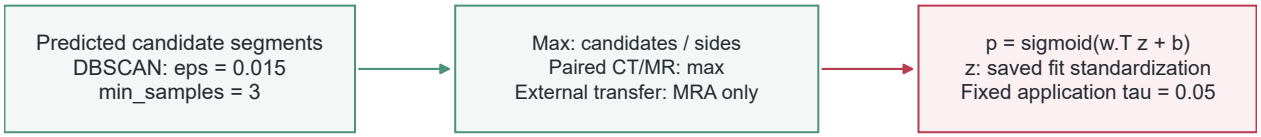

**d** Frozen hierarchical representation and saved readout




FullInformationPP shares native path-distance information across its information-matched controls.



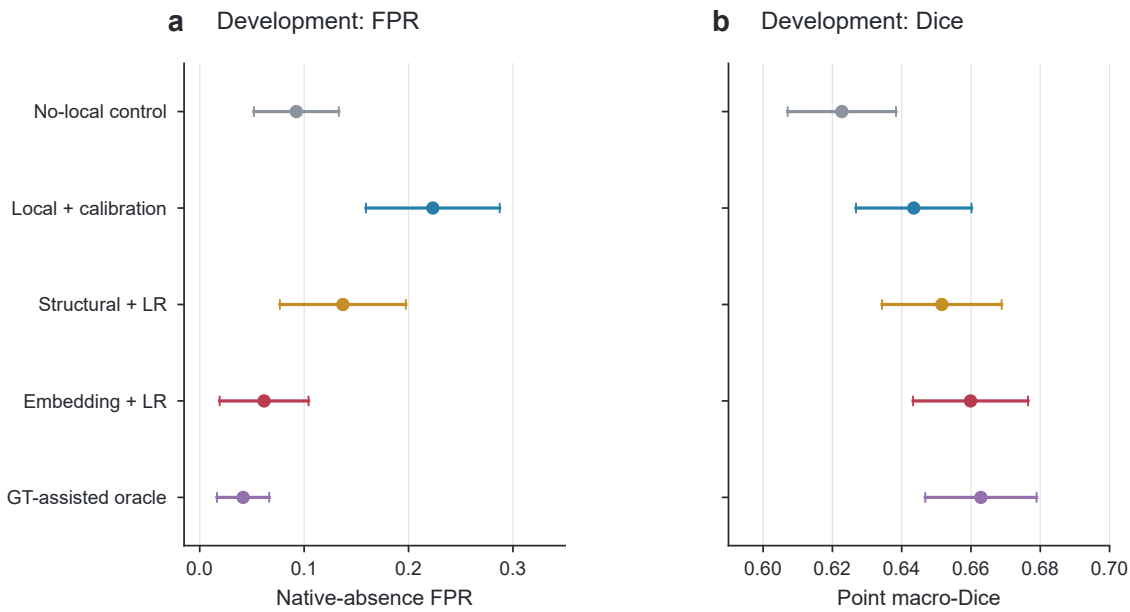

**e** Candidate-to-patient presence decision and execution



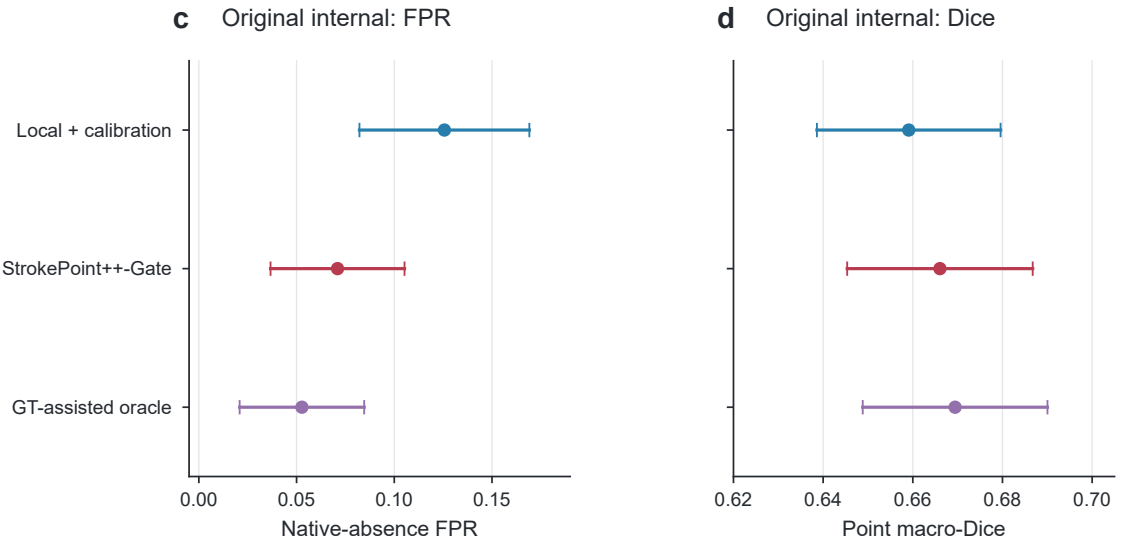

If presence evidence falls below the saved threshold:

block Pcom labels 8/9 or third-A2 label 15; reassign only affected argmax predictions.

Candidate generation and geometric neighborhoods are different operations. Spatial proximity is not vessel connectivity.



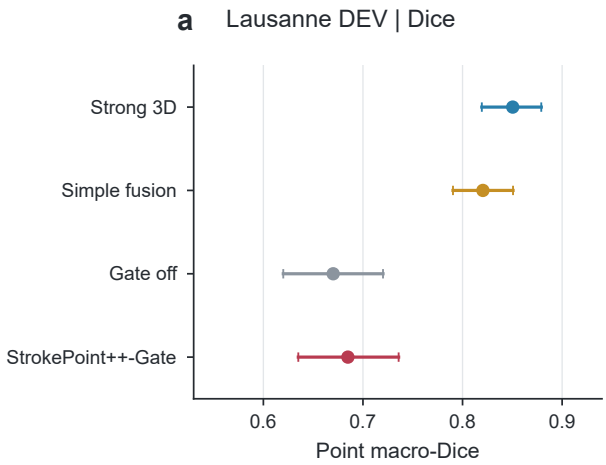

Exploratory development, 60 patients; FPR valid n = 58

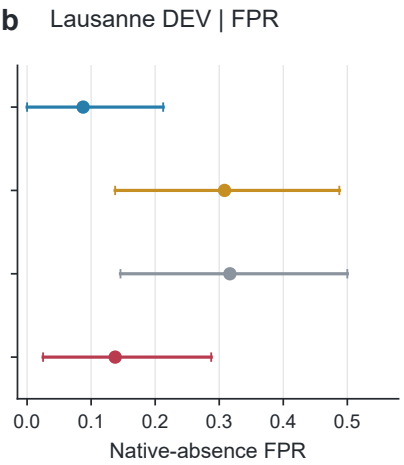


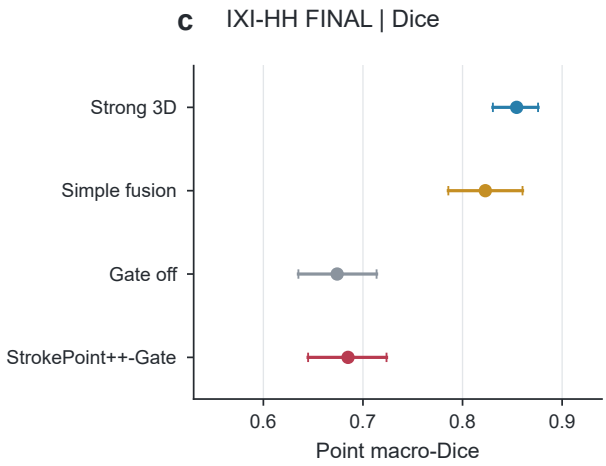

Internal held-out patients: Dice n = 65; FPR valid n = 64

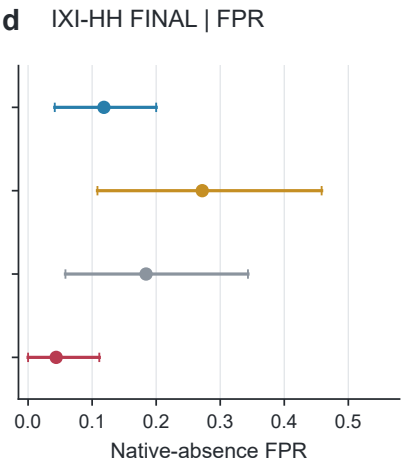
FPR 0.1257 -> 0.0710: 43.5% relative reduction

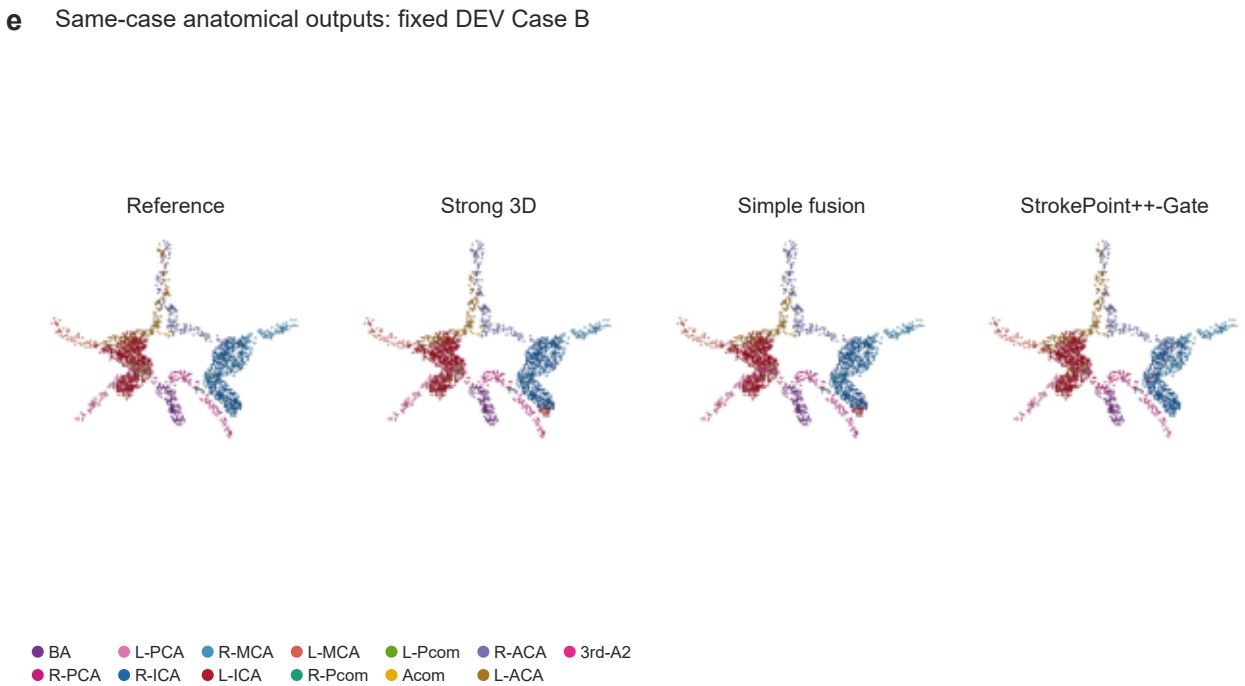
Original paired CT/MR application only. Historical audit limitations retained. Oracle uses target information and is not deployable.



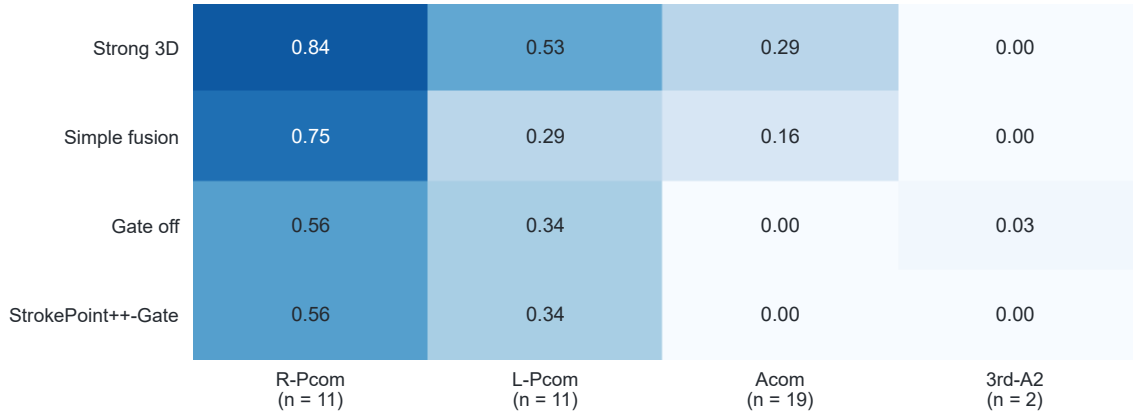



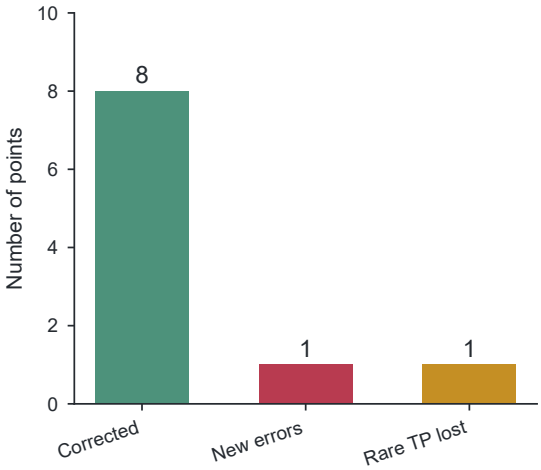

Cohorts remain separate. Fixed model checkpoints and thresholds; no FINAL tuning. Point macro-Dice, not full-volume Dice.

**a** FINAL rare-vessel point recall



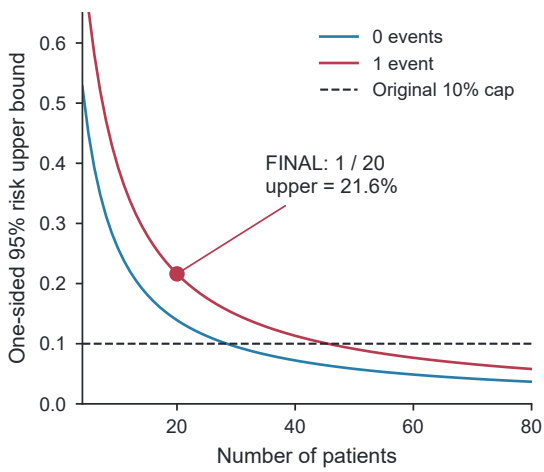

Native-positive patient support

**b** Gate edits versus its own output




31 modified points across 5 patients
Rare TP loss overlaps the new-error count.

**c** Patient-event risk precision




**d** What the present evidence establishes

**Application signal**

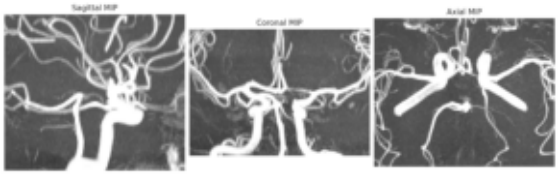
Gate reduces observed false identity presence, with a residual rare-point error.

**Unresolved target**

Acom identity and third-A2 recovery remain weak. Neither safety nor new-mechanism superiority passed.

No equivalence or safety claim from a zero-crossing interval. Acom negatives n = 1; third-A2 positives n = 2.

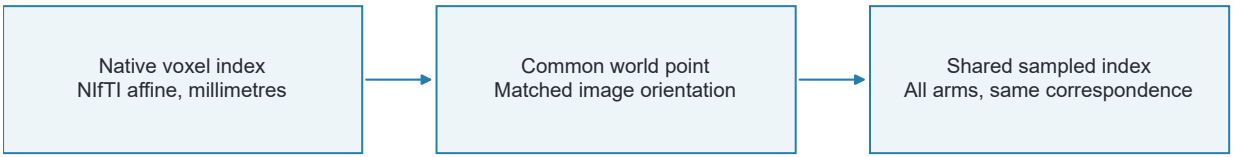
**a** Native MRA crop



**b** Input-only vessel surface



**c** Coordinate and input contract



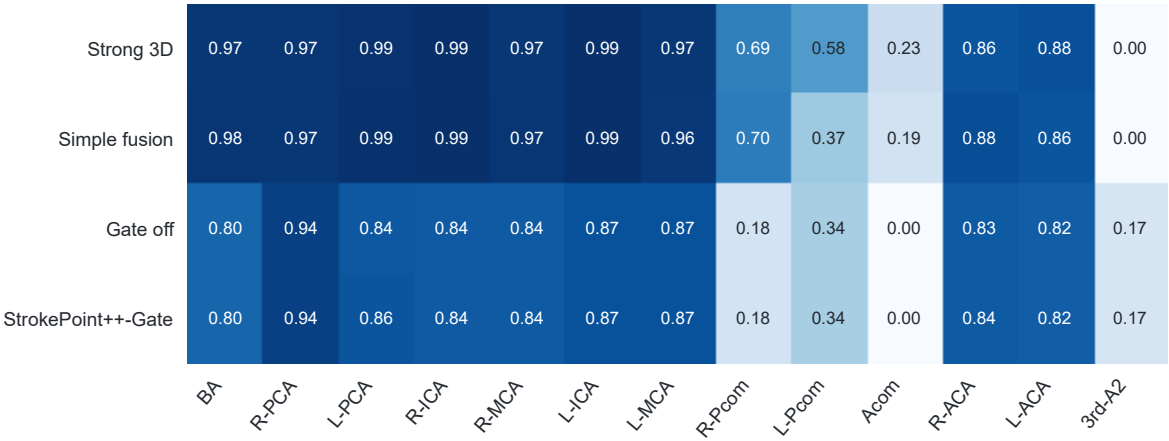

Only the binary union conditions preprocessing. Class identities remain evaluation targets.

**d** Sampling does not establish anatomical absence

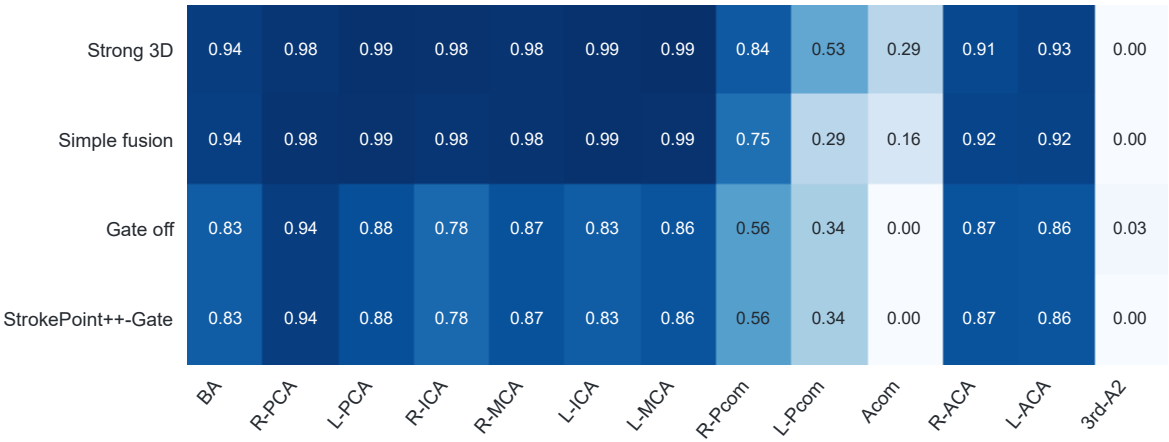



All examples are fixed DEV illustrations, not augmentation experiments or a clinical segmentation workflow.

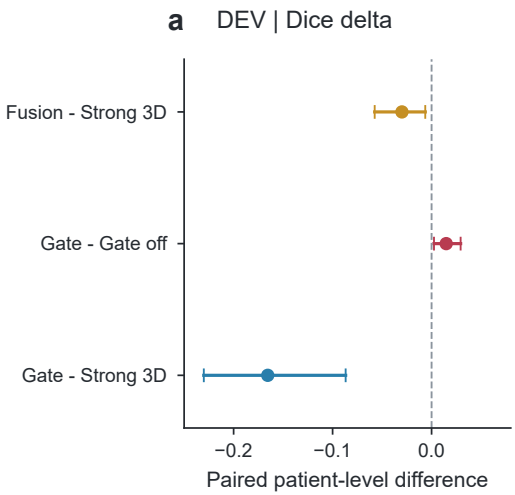
**a** DEV | full 13-class point recall

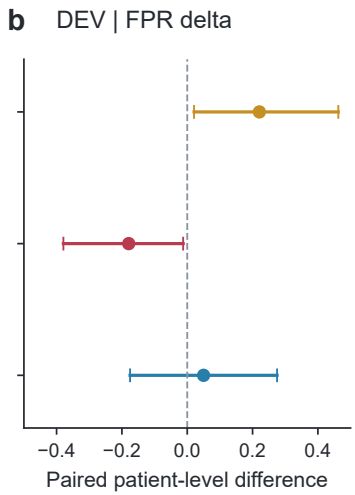


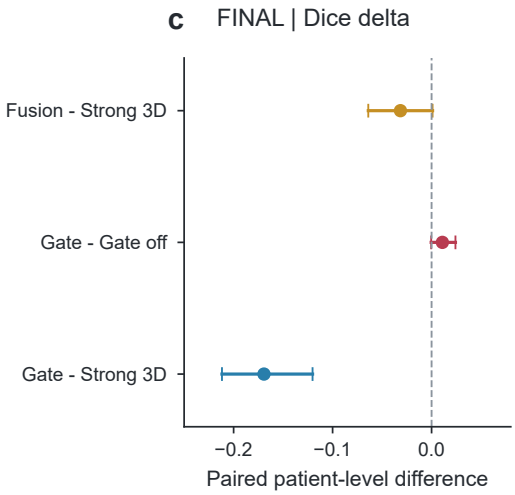

**b** FINAL | full 13-class point recall

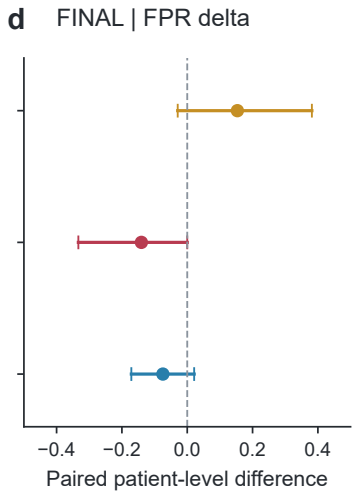



Canonical labels include third-A2 = 15. Empty-support intervals remain undefined rather than silently zero-filled.



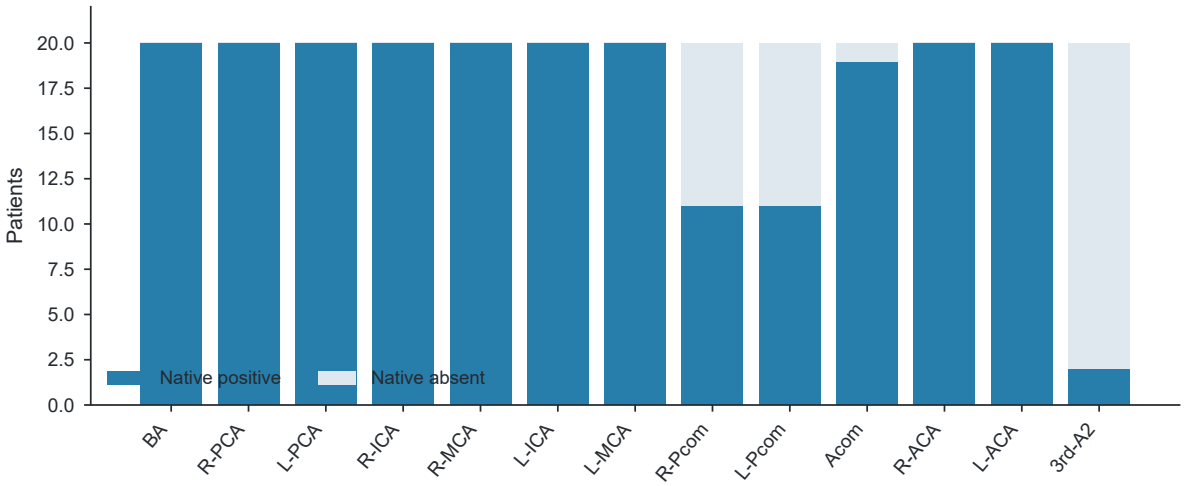




Original simultaneous family alpha = 0.05 / 3. Intervals include pairing; sign alone is not statistical superiority.

**a** FINAL native and sampled class support




**b** Sparse-support caveats

**Acom**

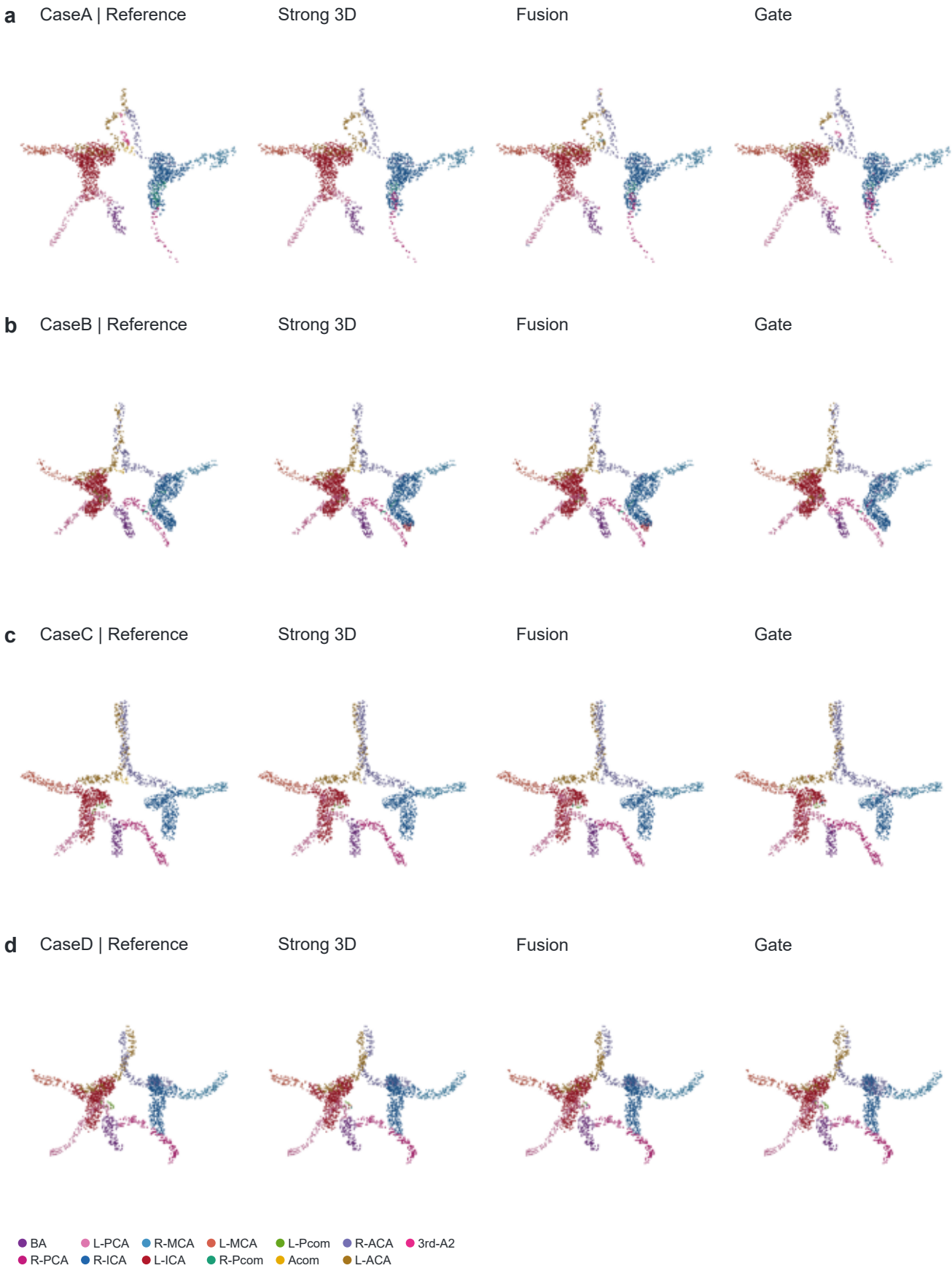
19 native-positive patients
1 native-negative patient
Absence-FPR uncertainty unresolved

**c** Metric denominators

**Third-A2**

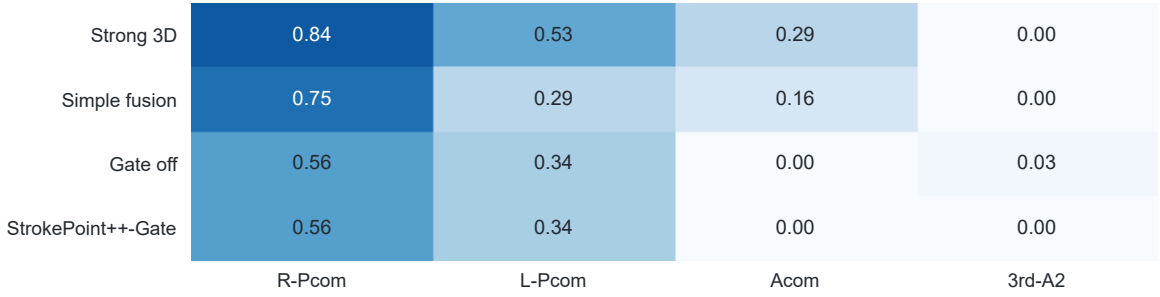
2 native-positive patients
63 sampled reference points
Point-recall interval undetermined

Overall Dice: n = 20. Overall absent FPR: n = 19. Patient-event risk: n = 20. These are different denominators.

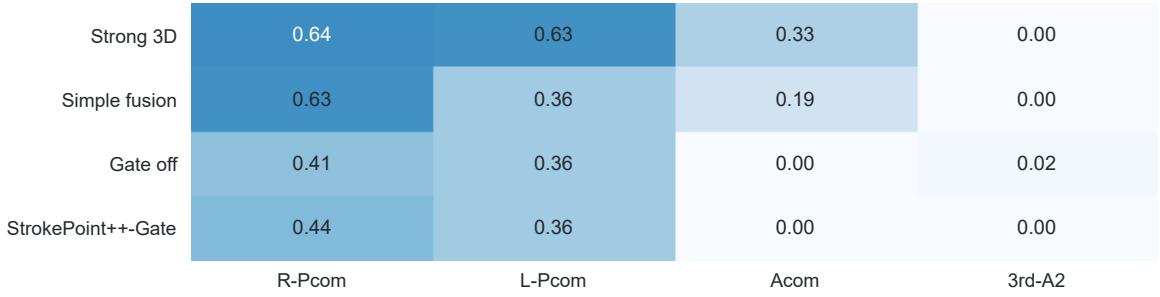



First four frozen Lausanne DEV cases. Identical within-case camera and point mapping; not selected for model quality.

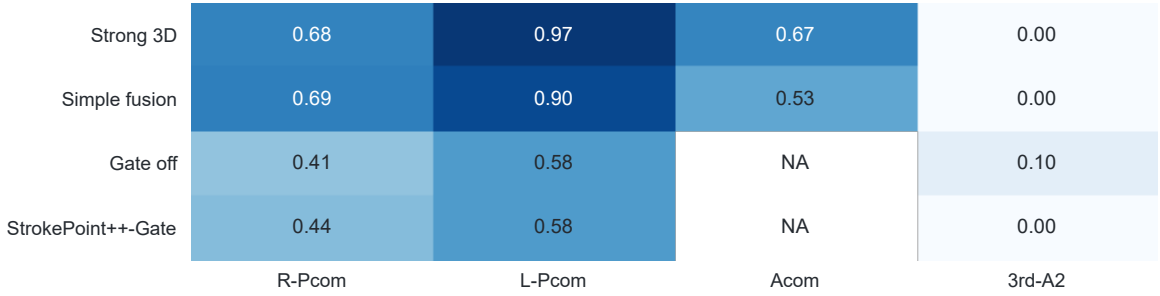
**a** FINAL rare classes | recall




**b** FINAL rare classes | Dice




**c** FINAL rare classes | precision




Means are descriptive with their own eligibility sets; see source JSON for CI status and exact denominators.

**a** Version and evidence separation

### Gate-AppYZ-v1

Original paired CT/MR application

Internal aggregate evidence; original Y/Z preserved

### Gate-MRA-R0v2

Fixed-recipe reconstruction

Single-MRA external transfer, not original LR states

### StrokePoint-3D

Raw MRA + strong 3D baseline

Short fixed pilot; full nnU-Net optimization not claimed

### StrokePoint-Fuse

Simple image / point fusion control

Same input comparison, no superiority established

### Research / archive

OT, routing, value and risk-head attempts

Negative / invalid / stopped versions remain separate